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Analysis of ground water in some rural areas of Kanpur region of Uttar Pradesh

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Abstract : Water is the biggest and chief natural resource, necessary for conservation for survival of living organisms on earth. It is the biggest necessity for life and in developing countries like India ground water is the most important source for drinking, irrigation, and industrial purposes. But unfortunately due to wide spread over use of harmful chemicals in agriculture, industrial effluents, pollution of rivers etc. ground water is getting increasingly contaminated with pollutants.

A laboratory study was conducted for the quality assessment of ground water in some selected rural areas of Kanpur. For the experimental purposes, six water samples were collected from different locations of the city (Three open well and three bore well). The results were analyzed with standard value prescribed. It was concluded that the sampling sites No. A to D showed all the physico-chemical parameters within the range of standard water quality as prescribed by (W.H.O.). The results of present study also concluded that most of the physico- chemical parameters are within permissible limits except samples No. E and F.

Keywords: Physico-chemical Analysis, World Health Organization (W.H.O.), Permissible limits, Industrial effluents, Physico Chemical Parameters.

INTRODUCTION

Ground water is ultimate, suitable water resource for living organisms on earth because it is a balanced concentration of the salts and minerals for human consumption. In another words water is the biggest necessity for life. For the survival in the nature, water is very essential factor for the living organisms. Drinking water is obtained from a variety of sources, depending on availability of surface water like lakes, rivers, reservoirs, ponds and ground water (Baig *et al.,* 2009; Nagpal *et al.,* 2004).[1, 2]

In India, most of the population is totally dependent for their survival on ground water because it is the main source for drinking. In many developing countries including India, ground water is the most important source for drinking, domestic and irrigation purposes. People in city as well as rural areas are dependent on ground water for their domestic and occupational requirements. It is well known fact, that ground water is comparatively much better, clean and free from pollution in comparison of surface water, but in city and rural areas discharge of industrial effluents, solid waste dump and sewage caused the ground water got polluted by the many anthropogenic activities of human beings and created so many health problems. In present situation, over burden of population, unplanned way of urbanization and dumping of polluted water at inappropriate places caused the discharge of harmful compounds to the ground water and it caused serious problems for human beings. (Todd, 1980; Raja and Sharmila, 2002; Mahanta *et al.* 2004)[3,4,5].

The physico-chemical characters of water can be changed due to presence of different pollutants and nutrients through different sources like, industrial effluents from tanneries and leather factories, sewage and agricultural by products (Olimax and Sikorska, 1975)[6]. It is observed, that these effluents and sewage water are not treated and purified before their disposal, the ground water

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can be very dangerous and harmful for human consumption as well as for domestic uses. Now-a-days, over burden of population, dumping of polluted water in an open areas, and unplanned way of urbanization caused the discharged of harmful toxic chemicals in the ground water and it caused serious problems for human beings (Chilton, 1996).[7]

It is well known fact, that among the various sources of water, ground water is very safest and easily approachable source for drinking and domestic purpose but quality of ground water is influenced by many factors, which are responsible for the impurity of ground water in the natural conditions (Forstner and Wittman, 1981; Tiwari and Dhar, 1994; Canter and Sobatini, 1995 and Singh *et al*. 2009).[8,9,10,11] The importance of ground water for survival of human society can not be ignored. In India, specially in Uttar Pradesh, there are many thickly populated cities where more than 80% population are dependant on ground water for drinking and other domestic purpose (Ramachandraian, 2004)[12].

The objective of the present study is to assess the quality of ground water for drinking purpose of same selected rural areas of Kanpur Region of Uttar Pradesh. The investigation is also a possible attempt to carryout to study, the physico-chemical analysis of drinking water on the basis of their physico-chemical parameters of experimental samples in the laboratory analysis.

MATERIALS AND METHODS

STUDY AREA :

The experiment was conducted in the Department of Zoology at D.A.V. College, Kanpur. In the city, the hottest months are April to June. In these months, the temperature varies from 28°C to 44°C in summers, while the coldest months are December and January. The average annual rainfall in the city, between 900mm to 1450mm in the rainy season. The monsoon months are mid June to mid September. During the post mansoon period, the rate of humidity is very high. The quality of ground water in the study area is badly affected by the industrialization in the city. Due to over burden of population, improper drainage system, and many anthropogenic activities by human beings, the quality of ground water is badly affected in the study area.

PREPARATION OF WATER SAMPLES

Ground water samples were collected in the month of August, 2014. These samples were collected from three open well (Depth 40-70 feet). Similarly other water samples were collected from three bore well (Depth 175-225 feet). Open well samples and bore well samples were coded OW-S and BW-S respectively. The water samples were collected in sterilized screw-capped plastic canes of two liters as per standard procedure of collection prescribed. Just after the collection, the samples were stored at low temperature (5°C) to prevent vaporization and biodegradation before their physico-chemical analysis. Water samples from six sampling sites (three open well and three bore well) were collected at different places in the rural areas of the Kanpur Region. The sampling location and their sources are shown in the Table-I.

Table – I: The sampling location and their sources

Sample No.	Sampling Location	Source
OW-S-(A)	Kalyanpur-Bithoor Road	Open Well
OW-S-(B)	Rawatpur Gaon	Open Well
OW-S-(C)	Jajmau Industrial Area	Open Well
BW-S-(D)	Kalyanpur-Bithoor Road	Bore Well
BW-S-(E)	Rawatpur Gaon	Bore Well
BW-S-(F)	Jajmau Industrial Area	Bore Well

Quality Assessment of Ground Water in some selected rural areas of Kanpur Region :

The collected water samples were physico-chemically analyzed in an ideal laboratory conditions. Total fifteen physico-chemical parameters are analyzed. These parameters are Colour, Odour, pH-value, Temperature, Electrical Conductivity, Total Dissolved Solids, Total Hardness, Alkalinity, Calcium, Chloride, Nitrate, Turbidity, Sodium, Potassium and Magnesium. The physico-chemical parameters were estimated as standard methods (A.P.H.A.,1998)[13]. To ensure accurate, analysis was done in triplicates and for the results, mean value was taken for consideration. The result were compared with the prescribed guideline given by W.H.O. (1997) and Indian Standard (I.S.105500).

RESULTS AND DISCUSSION

The physico-chemical analysis, to test and assess, the water quality of ground water in various places has been carried out for the results. The status of water quality of ground water in same rural areas of Kanpur Region at different places is presented in Table-II.

pH Value :

The term pH is used to express, the intensity of alkalinity or acid condition in a solution. The pH values of water samples varied upto 7.2 to 8.1. The pH Value in the water samples within the prescribed limit. Although pH is not harmful as such but if its value is more than 11, it can cause eye irritation and skin disorder (Khan *et al.,* 2013)[14]. Most of the water samples are slightly alkaline due to the presence of carbonates and bi-carboates.

Temperature :

The temperature of water collected from the various sites of the sampling area was in between 22°C to 28°C. Temperature of water above 30°C is harmful and unfit for public use (Zajic, 1971)[15].

Electrical Conductivity (E.C.) :

Electrical conductivity is a measurement of water capacity to carry electric current. The (E.C.) values were recorded in between 1400 as prescribed by W.H.O. The E.C. range were 425 micromoles/cm to 2665 micromoles/cm in the sampling sites (B) and (F), because very high amount of dissolved inorganic substances are dissolved in ionized form.

Total Dissolved Solids (T.D.S.) :

The salinity of ground water is denoted by the Total Dissolved Solids (T.D.S.). The sampling sites (C), (E) and (F), higher range of T.D.S. recoded. It is obvious, that water containing more than 500 mg/L of T.D.S. is not suitable for drinking purpose.

Total Hardness (T.H.) :

According to Trivedi and Goel (1986)[16], total hardness is the property of water which shows the leather formation with soap and it also increased, the boiling point of water. The hardness values were recorded, which ranges from 185mg/L to 870mg/L. It is also recorded that in sample No. (B), (E) and (F) the hardness were higher than the prescribed limit.

Alkalinity :

Alkalinity is most important chemical parameter of quality assessment of water. It is a capacity to neutralize a strong acid into its normal form due to the presence of carbonate, bi-carbonate and hydroxide compound of Sodium, Potassium, and calcium. Generally, the alkalinity of water itself is not harmful, but large amount of alkalinity have bitter taste of water. The values of total alkalinity of water samples were recorded 175 mg/L to 265 mg/L.

Calcium (Ca^{++}) and Magnesium (Mg^{++}) :

It is well known fact, that the presence of calcium and magnesium in sample is directly related to the hardness of water. The presence of Calcium ranged between 35.2 mg/L to 128.4 mg/L found below the prescribed normal range.

The presence of Magnesium content in water samples were collected from the various sites ranging from 38.2 mg/L to 148.6 mg/L within the normal limit.

Chlorides (Cl^-) :

In the present study, chloride concentration were recorded 130 mg/L to 310 mg/L. Higher range of chloride concentration were recorded in the industrial area sites 310 mg/L and 280 mg/L i.e., sampling sites (C) and (F) due to discharged sewage near the sampling sites.

Nitrate (NO_3) :

The presence of Nitrate is very important for suitability for drinking purpose. In the sampling sites, the nitrate contents of the samples were present within the permissible limits.

Turbidity :

The turbidity in the ground water varied from 0.1 to 4.7 NTU. This range is below and normal value found within the limits prescribed by IS-10500-91.

Sodium (Na^+) :

The presence of sodium were recorded between 165.8 mg/L to 375.3 mg/L. But sample (A), (B) and (C) showed higher value than the prescribed limit as suggested by W.H.O.

Potassium (K^+) :

In fresh water, the chief source of Potassium by the mineralization of rocks, but due to disposal of waste water, the quantities of Potassium (K^+) increase. The presence of Potassium in the water samples varied from place to place as presented in Table-II.

In general, the water quality of ground water in rural areas of Kanpur City is not totally fit and suitable for drinking and domestic purpose because, many chemical parameters like Electrical Conductivity, pH Value, Alkalinity of Water, Total Hardness and presence of Calcium, Magnesium, Sodium, Potassium, Chloride and Nitrates ranges over then prescribed limits. Most of parameters in various samples fluctuate due to seasonal variations. Present study indicates that, the extent of pollution due to industrial discharge, urbanization, and other anthropogenic activities by human interventions in the ground water quality. It is also concluded that, water quality of ground water in Kanpur is not totally fit and suitable and causes so many health problems in human beings.

Table-II: showing the physico-chemical analysis of water sample A, B, C, D, E & F from different localities of kanpur city

S. No.	Physico-Chemical Parameters	Sampling Sites						WHO 1997	ISI 10500 -91
		A	B	C	D	E	F		
1.	Colour	Colour Less	Colour Less	Colour Less	Colour Less	Colour Less	Colour Less	-	-
2.	Odour	Odour Less	Odour Less	Odour Less	Odour Less	Odour Less	Odour Less	-	-
3.	pH Value	7.3	7.2	7.6	8.0	7.8	8.1	7.0-8.5	6.5-8.5
4.	Temperature	24°C	22°C	24°C	26°C	23°C	28°C	<30°C	-
5.	Electrical Conductivity (E.C.)	1325	425	1250	1175	1050	2665	1400	-
6.	Total Dissolved Solids (T.D.S.)	470	370	1750	520	1525	1225	1000	500
7.	Total Hardness (T.H.)	290	540	185	360	675	870	500	300
8.	Alkalinity	242	115	265	186	175	132	120	200
9.	Calcium (Ca^{++})	35.2	76.5	78.4	115.8	98.2	128.4	100	75
10.	Magnesium (Mg^{++})	45.80	148.60	38.20	75.60	60.10	146.0	150	30
11.	Chloride (Cl^-)	130	245	310	185	220	280	250	250
12.	Nitrate (NO_3^-)	18.4	17.4	35.6	40.2	38.7	28.7	45	100
13.	Turbidity	0.3	4.5	0.2	0.1	4.7	0.5	5.0	10
14.	Sodium (Na^+)	375.3	268.4	335.2	165.8	178.4	194.6	200	200
15.	Potassium (K^+)	4.838	0.425	2.015	3.800	1.050	1.008	-	-

NOTE :

1. Most parameters are represented in mg/L except pH value.
2. Turbidity = N.T.U.
3. Electrical Conductivity (E.C.) = Micromho/cm.
4. World Health Organization (W.H.O.)
5. Indian Standard IS-105500-91

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